Elias Firisa

eliassf73.github.io github.com/EliasSf73 eliassfirisa@gmail.com

Research Interests: Generative modeling theory (flow-matching, diffusion), computational neuroscience, variational inference

Education

Korea Advanced Institute of Science and Technology (KAIST) Feb 2022 – Aug 30, 2026
B.S. in Brain & Cognitive Sciences, Minor in Industrial & Systems Engineering Daejeon, South Korea

Selected Coursework: Diffusion & Flow Models, Statistical Machine Learning, Probabilistic Machine Learning, Data Science, Stochastic Modeling, Theoretical Neuroscience, Signals & Systems, Signal Processing, Probability & Statistics, Engineering Statistics, Linear Algebra, Data Structures, Discrete Mathematics

Preprints & Writing

Elias Firisa. *Probabilistic Invertibility of Rectified Flows Beyond Global Monotonicity.* Preprint, 2026. *[PDF]*

- Developed a generic-in-time invertibility theory for rectified flow maps via a transversality argument, replacing the standard global monotonicity assumption with an almost-sure injectivity guarantee under Gaussian initialization.

Diffusion and Flow Models, from Scratch. Open technical notes, 2026. *[Site]*

- A self-contained 45-topic syllabus from probability foundations to guidance, each topic with intuition, derivation, and working code, building the chain DDPM → score matching → SDE → probability-flow ODE → flow matching.

Elias Firisa. *Least Squares and Likelihood in Linear Regression.* Blog post, 2025. *[Post]*

Research Experience

Generative Modeling Research Intern — *U-AIM Lab, KAIST* *Codebase* Sept 2025 – Aug 2026

- Translated SDE/ODE formulations of diffusion and flow-matching processes into a modular PyTorch framework, reproducing theoretical convergence bounds for training dynamics and noise scheduling.
- Investigated the impact of variance-preserving schedules and guidance terms on the smoothness of the probability flow ODE, analyzing trade-offs between sample fidelity and solver step-size.
- Extended this framework with consistency-style objectives to study how flow-matching theory influences **latent geometry**, specifically the relationship between vector field curvature and generalization.

Research Intern — *Decision Brain Dynamics Lab, KAIST* Dec 2024 – June 2025

- Processed multichannel frontal EEG signals to extract spectral biomarkers (specifically Alpha-Beta power asymmetry) indicative of acute stress responses.
- Developed machine learning pipelines to classify stress states from physiological data, analyzing feature importance to identify robust neural indicators across different subjects.

Research Intern — *Leeds WormLab, University of Leeds* June 2024 – Aug 2024

- Validated the efficacy of simple linear baselines (DLinear) on nonlinear chaotic systems (Duffing oscillator), characterizing generalization limits across stable versus chaotic attractor regimes.
- Extended this framework to *C. elegans* neural recordings, investigating whether simple linear approximations could recover low-dimensional manifolds governing high-dimensional population activity.

Undergraduate Researcher — *Brain-Machine Intelligence Lab, KAIST* *Codebase* Feb 2024 – Nov 2024

- Investigated predictive-coding–inspired objectives within Variational Autoencoders (VAEs) to model hierarchical inference, motivated by the Bayesian Brain hypothesis.
- Conducted rigorous disentanglement analysis on the CelebA dataset, utilizing latent traversals and directional edits to quantify how well the model separated distinct semantic factors (pose, lighting).

Industry Experience

AI Engineer — *SAI Technologies* Sep 2025 – Present
Part-time during studies (Sep 2025 – Aug 2026), full-time since Sep 2026

- **Behavior Detection System:** Lead the computer-vision work for a camera product that detects a specific human behavior in surveillance video: object detection, multi-object tracking, and event logic over time.
- **Edge Deployment:** Deploy models on edge devices rather than servers (camera NPUs, Hailo accelerators, Raspberry Pi 5, NVIDIA Jetson): ONNX export, 8-bit quantization and calibration, and accuracy validation after compilation.

- **Interaction Tracking:** Implemented real-time multi-object tracking pipelines (YOLOv8, ByteTrack) to analyze temporal human–object interaction dynamics from continuous video streams.

ML Engineer Intern — *MolpaxBio* June 2025 – Sep 2025

- **3D Reconstruction Pipeline:** Developed a deep-learning framework to synthesize textured 3D face meshes from multimodal inputs (volumetric CT scans and 2D images) for surgical simulation.
- **Parametric Modeling:** Utilized FLAME topology to infer latent shape and expression parameters, ensuring topological consistency (watertight meshes) for downstream AI-driven manipulation.

Selected Technical Projects

Variance-Preserving Diffusion with Few-Step Sampling *[Code]*

- Diffusion model written from scratch in PyTorch with custom noise schedules; compared consistency-model and ReFlow-style shortcuts by FID at 1, 2, and 4 function evaluations.

Deep Learning & Computational Modelling 2023–2024
[Math-Modeling Code] · *[Neuro-Sim Code]*

- **Optimization of Lotka–Volterra Dynamics:** Estimated nonlinear ODE parameters by minimizing model-data error with grid search, Nelder–Mead optimization, and simulated annealing; compared convergence behavior across methods.
- **Machine Learning & Generative Modeling Workflows:** Implemented reproducible training and evaluation workflows covering denoising score matching, neural ODEs, feed-forward networks, logistic regression, regularization, and bias-variance analysis.
- **Neural Coding Theory:** Quantified information content in Poisson spike trains using entropy/Fano factor analysis and reconstructed visual receptive fields via reverse correlation (Gabor filters).

RAG System for Financial Q&A (NLP) 2024

- **Retrieval Pipeline:** Engineered a Retrieval-Augmented Generation (RAG) system, utilizing hybrid ranking (dense + sparse vectors) to ground LLM responses in unstructured financial data.

Teaching & Leadership

Undergraduate Tutor — *KAIST Education Support Team* Sept 2024 – Dec 2025

- **Academic Mentorship:** Selected to instruct undergraduates in quantitative disciplines (Linear Algebra, Probability) and core life sciences (General Biology, Systems Neuroscience), facilitating weekly review sessions and 1-on-1 coaching.

Community Leadership 2022 – June 2026

- **Head of Ethiopian Students Association (3 Years):** Led academic support and cultural integration initiatives for the Ethiopian student community at KAIST.
- **Founding Member (ASCEND KAIST):** Established and managed the Futsal division, organizing semester-long leagues to foster student wellness and cross-cultural engagement.

Skills

Mathematical Foundations: Stochastic Calculus (SDEs/ODEs), Probabilistic Models, Linear Algebra
Machine Learning: Diffusion & Flow-Matching, Variational Inference, Object Detection & Tracking, UMAP/t-SNE
Neuroscience Methods: Neural Population Dynamics, EEG Spectral Analysis, Synaptic Plasticity (STDP)
Edge AI: ONNX export, 8-bit quantization and calibration, Hailo and NVIDIA Jetson toolchains, Raspberry Pi 5
Languages & Tools: Python (PyTorch, NumPy, Pandas, OpenCV), MATLAB, Git, Docker, Linux

Honors & Awards

1st Place — Web3 Blockchain Competition (Decentralized Marketplace Track) 2023
KAIST Alumni Scholarship (Awarded for Academic Excellence) 2024
INSEO Engineering Scholarship (External Merit Scholarship) 2023
KAIST Scholarship (Full Tuition & Monthly Stipend) 2022 – 2026